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MILITARY SPECIFICATION

LOCKS, COMBINATION FOR GENERAL SERVICES ADMINISTRATION APPROVED SECURITY CONTAINERS, VAULT DOORS, AND SAFE LOCKERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers commercially available combination locks bearing the Underwriters Laboratory (UL) label for Groups 1 and 1R as defined in UL 768 (see 3.1).

1.2 Classification. Combination locks shall be of the following styles, models, classes, types, and sizes. A Reference Identification Number (RIN) has been established. The RIN is made up of the document identifier and the RIN Code Numbers (see 6.2.1).

	RIN Code No.
Style I - Manipulation, force, and thermal resistant	1
Model HC - Hand change combination	A
Model KC - Key change combination	B
Class TR - Top reading	C
Class FR - Front reading	D
Type T - Tube type	T
Type NT - Non-tube type	N
Size SD - Small dial	S
Size LD - Large dial	L

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5340

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	RIN Code
Style II - Manipulation, force, thermal, and radiographic resistant	2
Model HC - Hand change combination	A
Model KC - Key change combination	B
Class TR - Top reading	C
Class FR - Front reading	D
Type T - Tube type	T
Type NT - Non-tube type	N
Size SD - Small dial	S
Size LD - Large dial	L

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-392	- Plastic Molding Material, Acetal, Injection and Extrusion.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipping and Storage.
MIL-STD 167-1 (Ships)	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-889	- Dissimilar Metals.

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MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM B36 - Brass Plate, Sheet, Strip and Rolled Bar.
ASTM B86 - Zinc-Alloy Die Castings.
ASTM B117 - Salt Spray (Fog) Testing.
ASTM B159 - Phosphor Bronze Wire.
ASTM B197 - Copper-Beryllium Alloy Wire.
ASTM A276 - Stainless and Heat-Resisting Steel Bars and Shapes.
ASTM B584 - Copper Alloy Sand Castings for General Application.
ASTM D2436 - Forced Convention Laboratory Ovens for Electrical Insulation.
ASTM D4066 - Nylon Injection and Extrusion Materials.
ASTM D3955 - Polycarbonate (PC) unfilled and reinforced materials.

(Application for copies should be addresssed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

Aerospace Material Specification 5112 - Carbon Steel Wire (Spring Quality Music Wire).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 768 - Combination Locks.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The locks shall be a dial operated, mechanical, three or more wheel combination lock bolt type, and shall, as a minimum, conform to and bear the acceptance label for UL Group 1 or Group 1R as specified in UL 768 and meet all the criteria as specified herein. The UL label is only a prerequisite for consideration and shall not, in itself, constitute final acceptance or selection of the lock.

3.2 Lock samples.

3.2.1 Bid samples. Unless otherwise specified (see 6.2.), each bidder shall furnish, with bid, 14 samples of the combination lock with dial, dial ring, and change keys (if used) they propose to furnish under the contract for inspection as specified in 4.3.1. One intact unit of the selected product will be retained by the testing agency and one intact unit will be returned to the selected bidder to be used as a guide in manufacturing the first article and production quantity (see 3.2.2). Locks shall meet the requirements of 3.1 prior to being selected as a bid sample or first article sample.

3.2.2 First article. Unless otherwise specified (see 6.2), 8 locks with change keys shall be furnished for first article testing and approval (see 4.3 and 6.3). If the requirement for bid samples is waived, 14 locks with change keys shall be furnished for the first article testing and approval. The first article units shall be selected at random from the first 100 standard production units or from the contractor's current inventory. The contractor shall provide with the first article sample a certificate of compliance certifying that the samples conform to the requirements and bear the UL label for Group 1 or Group 1R locks of UL 768 as specified in the contract. If approved, one intact unit of the first article submission will be forwarded to the cognizant quality assurance representative and one intact unit shall be returned to the contractor to serve as the manufacturing standard. No deviation from the manufacturing standard shall be acceptable without formal written approval of the contracting officer in advance.

3.3 Materials. When specific materials are required, all parts shall meet the chemical and physical properties of the material specifications indicated in table I.

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TABLE I. Material.

Part	Material
Case, cover, dial and dial ring	Copper alloy sand casting ASTM B584 copper alloy UNS C85700; or zinc alloy die casting ASTM B86 UNS Z33521, or bronze of equal physical properties.
Combination lock bolt	Stainless steel ASTM A276 UNS S30400; or brass ASTM B36 copper alloy UNS C26800.
Spindle	Corrosion-resisting steel UNS S30200 or S30400 ASTM B276, or brass (copper alloy) UNSC C26800 of ASTM B36.
Springs	Phosphor bronze ASTM B159 copper alloy UNS C51000 temper H08; or copper-beryllium alloy ASTM B197 copper alloy UNS C17200; or stainless steel ASTM A313 type 302 or equal or carbon steel AMS 5112.
Wheels 1/ tension washers spacing washers, drive cam and lever	Brass ASTM B36 copper alloy UNS C26800; or stainless steel ASTM A276 UNS S30200 or S30400; or bronze of equal mechanical properties; or nylon ASTM D4066 PA112 or acetal plastic L-P-392, type I; or polycarbonate of ASTM D3935; or aluminum 2024-T3 with alidne 1200 aluminum chromate coating.
Screws	Brass or steel, cadmium or nickel plated.

1/ On style II locks, the wheels shall be made of nylon plastic in accordance with ASTM D4066 PA112, or polycarbonate of ASTM D3935, or acetal plastic in accordance with L-P-392, type I. The center of the wheels may be brass or zinc alloy. For style II locks, the center insert for the wheels may be nylon, acetal plastic, or metal such as zinc alloy ASTM B86 UNS Z35530 is acceptable. A stabilizing or lubricating agent may be added to the nylon or acetal plastic. Locks fitted with synthetic, plastic or otherwise nonmetal wheels shall function freely after exposure to plus 150 degrees (°) Fahrenheit (F) (65° Celsius (C)) for 112 hours and after exposure to 100 percent humidity at plus 80°F (27°C) for 112 hours.

3.3.1 Material deterioration and control. The lock shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

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3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, must either be plated or compatible to prevent operationally destructive corrosion as verified by the Salt Spray Test (see 4.5.2.5).

3.3.1.2 Identification of materials and finishes. When specified (see 6.2), the contractor shall identify the specific material, material finish, or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which has been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The parts incorporated in the locks may be newly fabricated from recovered materials to the maximum extent practicable, provided the locks produced meet all other requirements of this specification. Used, rebuild, or remanufactured parts shall not be incorporated in the locks.

3.4 Design. The lock shall be constructed to give maximum security. The lock shall have not less than a three wheel pack assembly and a driver cam. The opening tolerance shall not exceed one dial division on either side of the designated combination setting. Dials shall be top or front reading as specified in the contract. Locks shall be hand change or key change combination as specified in the contract. Dust covers shall be provided as specified in the contract.

3.4.1 Hand change locks. Hand change locks shall be designed to be readily combination code changeable by hand without the use of special tools or keys. Changing the combination shall require a minimum amount of disassembly and reassembly. The wheel-pack assembly (may include the back cover plate) shall be removable as a subassembly for servicing outside the lock case. Removal of the back cover shall not require removal of more than two screws. Wheels shall be clearly indexed with numbers that are easily read. Flies shall be attached to the wheels to prevent separation. The wheel post retainer shall be a "C" ring or similar retainer which is removable by hand. Hand change locks shall require no replacement of components during combination change. Lock design shall minimize the possibility of improper orientation of components during reassembly.

3.4.2 Key change locks. Key change locks shall be designed to be readily combination code changeable by the use of a special key or tool. The combination change shall not require disassembly of the lock.

3.5 Security. Style I and II locks shall resist opening by expert or professional manipulation for 20 man-hours, and forced entry for 10 minutes. Manipulation includes electronic enhancement of indicators obtained by feel, sight, or sound. The resistance of 3.5 shall be accomplished by the lock(s) after being turned at 600 revolutions per minute (rpm) clockwise for 4 hours and counterclockwise for 4 hours. The forced entry attack is restricted to using a limited set of tools (see 4.5.2.7). After the 8 hour break-in period,

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the lock shall have a torque that is less than twice the starting value and a combination that has not changed by more than 1 division. Further, the style II lock shall resist radiological attack with radioactive isotopes, such as Cobalt 60, not to exceed 10 curies at 30 inches (762 mm) for 20 hours. After the 8 hour break-in period, the lock shall also be normally functional.

3.6 Construction.

3.6.1 Case and cover. The case and cover shall be a maximum of 2.40 by 3.40 by 1.40 inches (60.96 by 86.36 by 35.56 mm). The cover shall be held in place by not less than two screws each fitted with a lock washer, or other means of securing the screw. The lock shall be attached to the plate on the door by means of four, 0.25 inch diameter, (6.35 mm) 20 UNC-2A machine screws, located as shown in figure 1. These four machine screws shall be of sufficient length to extend 0.250 inch (6.35 mm) beyond the exterior of the lock case; these machine screws shall be furnished with the lock. The walls of the case shall be minimum 0.060 inch (1.524 mm) thick. The slot in which the lock bolt operates shall be cast or die cast as an integral part of the lock case. When sheet material is used, the slot shall be cut or stamped in the case. The portion of the bolt within the case shall be supported by this slot or other suitable means in the locked position. The bolt and case shall resist a force of 600 pounds (lbs) (2669 Newtons (N)) applied at a rate of 60 lbs (267N) per second (see figure 2).

3.6.2 Spindle. Unless otherwise specified (see 6.2), the lock shall be furnished with a spindle which can be cut to suit varying thicknesses of doors, drawers, and back plates not less than 4.5 inches (114.3 mm) in combined thickness. The spindle or cam shall be so designed to permit the lock to be opened on a final designated setting when installed in right hand, left hand, up or down positions.

3.6.3 Wheel-pack assembly. The wheel-pack combination shall be of the hand or key changing type, as specified (see 6.2) and shall be capable of 800,000 theoretical combination changes, and shall be actuated by a pin on the driver wheel which interlocks with the third or fourth wheel. The remaining wheels shall each actuate the next lower number wheel, that is, the third wheel shall actuate the second wheel and the second wheel shall actuate the first wheel. The wheel centers shall resist slippage when torqued to 5 inch-pound-force (lbf.in) (0.575 Newton metre (N.m)) in either direction.

3.6.3.1 Wheel-pack assembly torque. The locks shall maintain a wheel-pack assembly torque in the range of not less than 16 inch-ounce-force (ozf.in) (0.1136 N.m) and not more than 24 ozf.in (0.1704 N.m).

3.6.4 Drive cam and lever. The drive cam and lever shall be so arranged that no external forces may be applied to its operation or that its operation would cause wear which would prevent the lock from meeting requirements of 3.4 or 3.5.

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3.6.5 Lock bolt. The lock bolts cross-section shall be not less than a nominal 0.312 by 1.00 inch (7.92 by 25.4 mm) with a minimum projection from the case of 0.375 inch (9.55 mm) in the locked position, and flush to 0.062 inch (1.575 mm) in the unlocked position, and shall have not less than 0.312 inch (7.92 mm) inch throw.

3.6.6 Relocking trigger. The lock shall be fitted with a relocking trigger which will deadlock the bolt in the locked position if the drive Cam is moved from its normal working position, or the back cover is moved more than 0.10 in (2.54 mm) from its normal operating position. The relocking trigger shall substantially deadlock the bolt so that the relocker may not be overridden. The lock shall also be fitted with a thermal relocking trigger which will deadlock the bolt in the locked position when the temperatures within the lock reaches 165°F (91.6°C) or greater.

3.6.7 Dials and dial rings. As specified (see 6.2), a top or front reading dial shall be provided with suitable dial rings. "The top reading dial ring shall be 3.00 to 4.00 inches (76 to 101 mm) in diameter. The top reading dial shall be marked with 100 graduated divisions numbered from 0 to 90 in steps of 10, with each 5th division distinctively identified to facilitate easy reading. Black background with white markings or tarnish proof metalic background with black markings are preferred. Dial design shall ensure high security against casual observer being able to read dial combinations. Logos or nonessential markings shall not appear on the dial knob or other exposed viable surfaces. A durable dust cover shall be supplied to cover the dial for both dials. This cover must not interfere with normal dialing operation. The front reading dial shall have basically the same numbering and graduated divisions and colors as the top reading dial. The front reading dial diameter shall be approximately 2.4 in (61 mm) for the small dial and 2.9 in (74 mm) for the large dial as specified (see 6.2).

3.6.8 Tubes. When specified (see 6.2), the lock shall include a tube that fits over the spindle to provide protection for the spindle and works from insulation and other unwanted material in the works. The tube shall be metal and shall be adjustable by cutting to bridge the combined thickness of doors, insulations, backplates, etc., of not less than 4.5 inches (114.3 mm).

3.7 Operation. The wheel pack, driver cam, and other parts shall operate smoothly and quietly except for sounds produced by soundbaffle, for the lifetime of the lock, without the addition of anything but proper lubricants and without showing appreciable wear. The assembled lock shall be capable of withstanding 10,000 complete and continuous cycles of turning the dial at a speed not exceeding 48 rpm, without failure. One cycle consists of dialing the combination, retracting the bolt, throwing the bolt, and scrambling the combintaion. These locks shall withstand, without failure, 50 combination changes including three open and close operational verifications after each change. Dialing tolerance within the maximum acceptable limits of one division either side of the combination setting shall also be verified (see 3.5).

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3.7.1 Vibration. When specified (see 6.2), locks intended for severe duty (such as high frequency and high amplitude vibrations aboard ship) shall be subjected to type I environmental vibration tests of 4.5.2.8. Operation and security performance and tolerances shall remain within standards.

3.8 Marking. Each lock shall be permanently marked with the style, manufacturer's name or trademark on the case, coverplate, or backplate. The applicable UL label shall be attached.

3.9 Finish. All surfaces shall have a uniform finish of sufficient smoothness to accept the markings required.

3.10 Instructions. Instructions for dialing the combination to which the lock is set, as well as to determine the length of the spindle or tube to be cut-off to suit varied combined door and back plate thickness shall be furnished by the contractor to the Government as normally furnished in commercial practice for each look specified in the contract.

3.10.1 Key. When required, a key(s) for changing the combination numbers setting shall be furnished with each lock unit.

3.11 Workmanship. The locks shall be free of sharp edges, burrs, slivers, and any defects affecting the appearance, operation, or serviceability.

3.12 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied. Customary units are used in this specification, including a Soft Conversion by formula to Metric Units.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Bid sample inspection (see 4.3).
b. First article inspection (see 4.3).
c. Quality conformance inspection (see 4.4).
d. Inspection of packaging (see 4.6).

4.3 Bid sample and first article inspection.

4.3.1 Bid sample and first article examination. The bid sample and first article shall be examined as specified in 4.5.1 and tested as specified in 4.3.2 and 4.3.3. The presence of one or more defects shall be cause for rejection.

4.3.2 Bid sample tests. Bid samples shall be tested as specified in 4.5.2. Check starting torque before any tests are conducted. Failure of any test shall be cause for rejection.

4.3.3 First article tests. The first article shall be tested as specified in 4.5.2.1, 4.5.2.2, 4.5.2.3, and 4.5.2.4. Failure of any test shall be cause for rejection. Check starting torque before any tests are conducted. When bid submission is waived, verification of all requirements of this specification and performance of all tests of 4.5.2 are applicable for first article acceptance.

4.4 Quality conformance inspection.

4.4.1 Lot. A lot shall consist of all the locks of the same style offered for delivery at one time.

4.4.2 Sampling. Sampling of locks for examination and test shall be in accordance with MIL-STD 105, inspection level II for examination and level S-1 for test.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the defects specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.4 Test. Samples selected in accordance with 4.4.2 shall be tested as specified in the combination change portion of 4.5.2.2 and the wheel slippage test of 4.5.2.4. AQL shall be 2.5 percent defective.

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4.5 Inspection procedure.

4.5.1 Examination. The locks shall be examined for defects in accordance with table II.

TABLE II. Examination for defects.

Defect No.	Defect	Requirement Paragraph
101.	Materials not as specified.	3.3
102.	Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
103.	Dissimilar metals as defined in MIL-STD-889 are not (treated or plated to prevent corrosion.)	3.3.1.1
104.	Contractor does not have documentation available for identification of material, material finishes or treatment.	3.3.1.2
105.	Used, rebuilt or remanufactured components, pieces or parts incorporated in the locks.	3.3.2
105.1	Design not as specified	3.4, 3.4.1, 3.4.2
106.	Security of locks not as specified.	3.5
107.	Dimensions not as specified.	3.6.1 thru 3.12
108.	Wheels not as specified.	3.6.3
109.	Drive cam and lever not as specified.	3.6.4
110.	Lock bolt not as specified.	3.6.5
111.	Relocking trigger not as specified.	3.6.6
112.	Dial ring diameter not as specified.	3.6.7
113.	Dial not as specified.	3.6.7
114.	Dust cover not as specified.	3.6.7
115.	Identification markings, incorrect, illegible, or missing.	3.8
116.	Finish not as specified.	3.9
117.	Instructions not furnished, or not as specified.	3.10
118.	Workmanship not as specified.	3.11

4.5.2 Tests. Unless otherwise specified (see 6.2), sample units shall pass tests of 4.5.2.1 through 4.5.2.7 (see 4.3.3 and 4.4.4).

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4.5.2.1 Security. The lock in the locked condition shall have the dial turned at 600 rpm for a break-in period of not less than 8 hours, turned 4 hours in clockwise and 4 hours in counterclockwise direction. After the break-in period, an attempt shall be made to manipulate the lock, including the use of electronic enhancement of any indicators that may be found. The lock shall resist manipulation for 20 man-hours. Forced entry shall be resisted for 10 minutes. The ending torque shall be checked when finished and before proceeding to 4.5.2.2 to determine conformance to 3.5. The lock shall be checked for normal functioning after the 8 hour break-in period.

4.5.2.2 Operation. The lock shall withstand, without failure, after being worked 10,000 cycles (see 3.7), 50 combination changes including three open and close operational verifications after each change. Dialing tolerance within the maximum acceptance limits of one division either side of the combination setting shall also be verified.

4.5.2.3 Case and bolt strength. Mount the lock in its normal manner on a door so that the bolt extends at least .10 inch (2.54 mm) beyond the door edge as shown in figure 2. Apply a force against the bolt as shown in the figure at a rate of 60 lbf (267 N) per second until a minimum threshold of 600 lbf (2669 N) has been applied to the bolt. Any fracture, or bending of the bolt or case shall constitute failure of this test (see 3.6.1).

4.5.2.4 Wheel to wheel center slippage. Mount each wheel so that it remains stationary. Apply a torque of 5 lbf.in (0.575 N.m) to the wheel center (or insert) in a clockwise rotation and then a counterclockwise rotation. Any evidence of slippage at less than the required torque between the wheel center and the outer wheel shall constitute failure of this test (see 3.6.3).

4.5.2.5 Salt spray corrosion test. Subject the lock to the Salt Spray (Fog) test of ASTM B117 for 72 hours. Operate the lock at 24 hour intervals during the 72 hours. Following the exposure, the lock is allowed to dry for 24 hours in ambient temperature 77°F (25°C). After the drying period, operate the lock as intended on the combination to which it is set. Failure to operate as intended constitutes cause for rejection.

4.5.2.6 Polymeric materials (synthetics) tests.

4.5.2.6.1 Strength. The synthetics mechanical strength and impact resistance will be considered adequate for the intended purpose upon satisfactory performance during the security test and operation test of 4.5.2.1 and 4.5.2.2 and the forced entry and vibration tests of 4.5.2.7 and 4.5.2.8.

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4.5.2.6.2 Aging-stress distortion. The synthetic shall not warp to the extent that the intended function is impaired when the material is aged for 7 days in a circulating-air oven conforming to ASTM D2436 with temperature maintained 194°F ± 1.8°F (90°C ± 1.0°C). Exhaust ports of the oven shall be adjusted to achieve 200 or more complete fresh-air changes per hour. A substantial amount of fresh air is to be admitted continuously throughout the 7 day aging period. At least 6 representative samples shall be tested. Following the 7 day aging, the samples are removed, cooled at room temperature, and then checked for performance of the intended function by conducting the operation and wheel slippage test of 4.5.2.2 and 4.5.2.4.

4.5.2.6.3 Moisture absorption. The synthetics shall function as intended after this test procedure.

a. Condition 3 representative samples as described for aging-stress distortion test of 4.5.6.2.2.

b. Expose the samples for 10 days in a Chamber producing warm, moist air of relative humidity of 98 ± 2 percent and maintained at 140°F (60°C) dry bulb temperature. The apparatus for warm, humid air exposure is to consist of an insulated specimen chamber having a temperature controlled water reservoir with a pump, a spray chamber for humidifying the air, an air circulating fan, provision for heating the air, specimen supports, and the necessary means of control. Following the 10-day exposure, the samples are to be removed and permitted to cool. As soon as the samples reach room temperature, they are to be tested for intended operation by subjecting the samples to the operation and wheel slippage tests of 4.5.2.2 and 4.5.2.4.

4.5.2.7 Forced entry test. While installed on a GSA approved class 5 secruity container, the lock must resist forced entry attack for a minimum of 10 minutes. The tools used for the forced entry attack are limited to hand or battery powered tools not to exceed 25 pounds total weight with a maximum tool length of 24 inches for any one tool. Throwing the bolt to the open position without knowledge of combination in less than 10 minutes will constitute failure of this test.

4.5.2.8 Environmental vibration (type I). Conduct the type I environmental vibration test of MIL-STD-167-1 (Ships). Check for conformance to the operation, security and tolerance requirements of 3.7.1.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2. Sampling. Sampling for examination shall be in accordance with MIL-STD 105.

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4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the defects of table III. AQL shall be 2.5 percent defective except as noted (see 1/).

TABLE III. Defects for examination.

Defect No.	Defect	Level A	Level C
117. 1/	Unit containers not sealed as specified. They can be opened and resealed without evidence of tampering.	5.1.1	5.1.1
118.	Key and instruction sheet not in unit container with lock as specified.	5.1.2.1	5.1.3.1
119.	Unit container not sealed with reinforced tape as specified.	5.1.2.1	5.1.3.1
120.	Sealed bag enclosing the unit container for level A preservation not style 2 (transparent) as specified.	5.1.2.1	
121.	Improper quantity of locks placed in intermediate container.	5.1.2.2	5.1.3.2
122.	Intermediate container not as specified.	5.1.2.2	5.1.3.2
123.	Closure of intermediate container not as specified.	5.1.2.2	5.1.3.2
124.	Shipping container not as specified.	5.2.1	5.2.2
125.	Shipping weights exceed specified limitations.	5.2.1	5.2.2
126.	Marking not in accordance with MIL-STD-129.	5.3	5.3
127.	Marking not in accordance with contract or order.	5.3	5.3
128. 1/	Item description (Name/nomenclature), marked on unit container.	5.3	5.3
129. 1/	Unit containers not marked or labeled with special instructions as specified.	5.3.1	5.3.1

1/ The finding of one or more of defects 117, 128, or 129 shall be cause of rejecting the entire lot.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 General. Each lock, including the envelope containing the manufacturer's instruction sheets and, as applicable, the combination changing key (see 3.10), shall be preserved in an individual container that is sealed in such a manner that it cannot be opened and resealed without obvious indication of tampering.

5.1.2 Level A.

5.1.2.1 Unit container. Each lock, with envelope containing applicable instructions and key (see 3.10 and 5.1.1), shall be preserved in accordance with submethod 1C2 of MIL-P-116. The snug-fitting carton or box shall comply with PPP-B-566 or shall be the contractor's standard commercial snug-fitting carton or box. Closure of the carton or box shall be with reinforced tape. The sealed bag enclosing the carton or box shall be as specified for the submethod, except that the bag shall be style 2.

5.1.2.2 Intermediate container. Ten locks, each preserved in the unit container specified in 5.1.2.1, shall be placed in a close-fitting fiberboard box conforming to PPP-B-636, class-weather-resistant. Box closure shall be in accordance with method V of the appendix thereto.

5.1.3 Level C.

5.1.3.1 Unit container. Each lock, with envelope containing applicable instructions and key (see 3.10 and 5.1.1), shall be preserved in a close-fitting fiberboard folding or set-up carton or box, sealed with reinforced tape.

5.1.3.2 Intermediate container. Ten locks, each preserved in the unit container specified in 5.1.3.1, shall be placed in a close-fitting fiberboard box conforming to PPP-B-636, class domestic. Box closure shall be in accordance with method I of the appendix thereto.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Locks, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2, style optional. The gross weight of the box shall not exceed 200 lbs. Box closure and strapping shall be as specified in the applicable box specification or the appendix thereto, except that the strapping shall be flat and its finish shall be B.

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5.2.2 Level C. Locks, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-636, class domestic, either single wall or double wall. The quantity per box shall not exceed the applicable weight limitations specified therein and the box closure shall be method 1 of the appendix thereto.

5.3 Marking. Marking shall be specified in the contract or order and in accordance with MIL-STD-129 except that the item description (name/ nomenclature) shall be omitted from the unit containers.

5.3.1 Additional marking. Each unit container specified in 5.1, shall be marked with the following special instructions:

IMPORTANT

TO BE OPENED BY DESIGNATED

USER SECURITY PERSONNEL ONLY

The letters shall be 1/4 inch high minimum, color to be red or black, and shall be applied by marking the reinforced sealing tape (see 5.1.2.1 and 5.1.3.1) or by the application of pre-printed labels.

6. NOTES

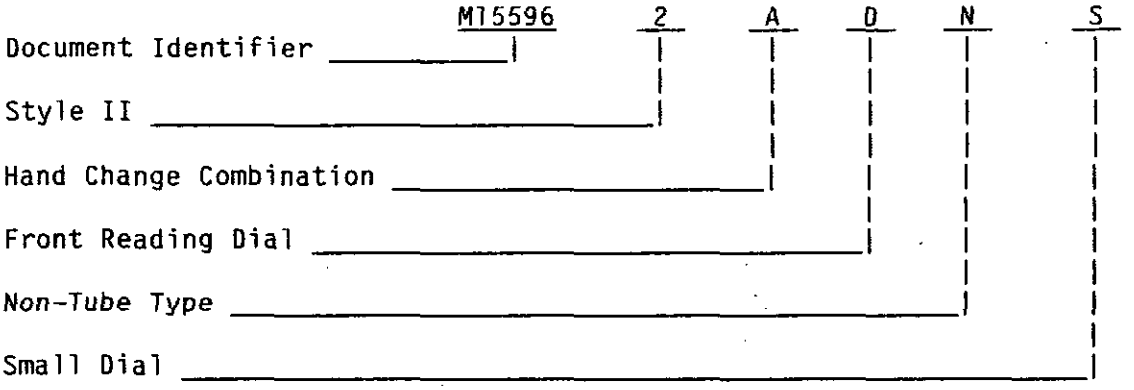
6.1 Intended use. Combination locks covered by this specification are intended primarily for use on safes, safe lockers, security containers, vault doors, and similar items.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Style, model, class, type and size required (see 1.2, 6.2.1, and 6.2.2).
c. Bid samples required and instructions for tests (see 3.2.1).
d. When a first article is required for inspection and approval, and the number of units required (see 3.2.2).
e. Identification of materials and finishes required (see 3.3.1.2).
f. Top or front reading dial and dust cover required (see 3.4).
g. Hand or key change combination required (see 3.4 and 3.6.3).
h. When spindle shall be other than as specified (see 3.6.2).
i. Size of front reading dial required (see 3.6.7).
j. When tube is required and combined thickness of material the tube must extend through (3.6.8).
k. Vibration test required (see 3.7.1).
l. Deviations from tests specified (see 4.3.3, 4.4.4, and 4.5.2).
m. Instructions for First Article tests (see 6.3 and 6.4).
n. Level of preservation and level of packing required (see 5.1 and 5.2).

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6.2.1 Reference identification number. An example of RIN make-up for a style II, hand change combination, front-reading, non-tube type with small dial:



6.2.2 Cross-reference. The cross-reference of previous and current edition classifications:

MIL-L-15596F	MIL-L-15596G
Style I	Style I
None	Model HC
None	Model KC
None	Class TR
None	Class FR
None	Type T
None	Type NT
None	Size SD
None	Size LD
Style II	Style II
None	Model HC
None	Model KC
None	Class TR
None	Class FR
None	Type T
None	Type NT
None	Size SD
None	Size LD

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model. The first article should consist of 8 or more unit(s). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.4 First article units. Four each required or 7 required if bid sample is waived. First article units should be sent to one or both of the following addresses for inspection and approval as specified in the contract.

a. Director
U.S. Army Intelligence Materiel Activity
ATTN: AMXI-M-D
Fort Meade, MD 20755-5315

b. Commander
Naval Civil Engineering Laboratory
ATTN: Code L56
Port Hueneme, CA 93043-5003

6.5 Subject term (key word) listing.

Combination lock
Container lock
Lock files
Lock, vault
Group 1 lock
Group 1R lock

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

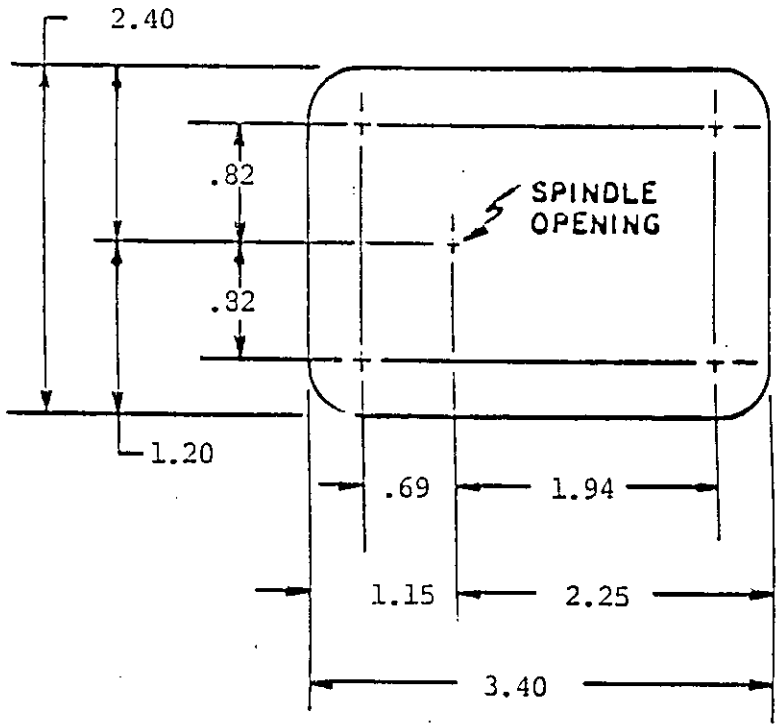
Review activities:
Army - AR, GL
DLA - IS

User activities:
Army - CE
Navy - MC SH

Preparing activity:
Navy - YD

(Project 5340-1819)

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NOTE: ALL DIMENSIONS ARE APPROXIMATE AND IN INCHES.

FIGURE 1. Schematic arrangement of bolt holes.

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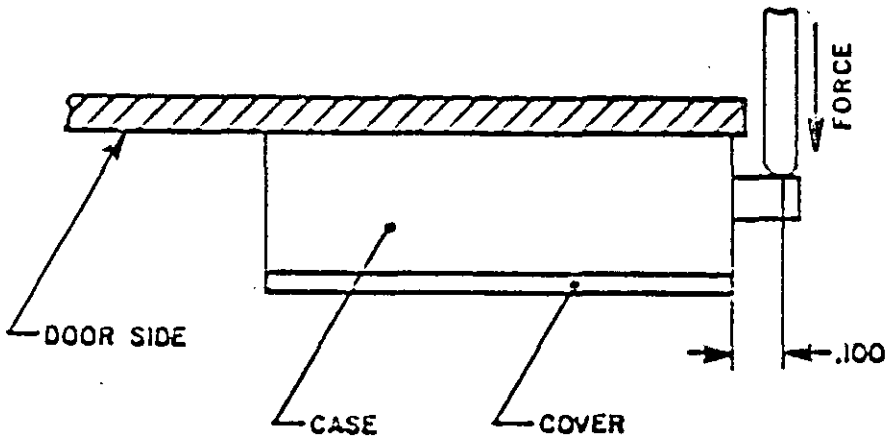


FIGURE 2. Case and Bolt Strength test.

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Commanding Officer (Code 156)
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-15596G	LOCKS, COMBINATION FOR GENERAL SERVICES, ADMINISTRATION APPROVED SECURITY CONTAINERS, VAULT DOORS, AND SAFE LOCKERS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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